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SUPERSEDING
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10 October 1978

MILITARY SPECIFICATION

FOOTWEAR COVER, CHEMICAL PROTECTIVE (OVERBOOTS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a footwear cover which provides protection against chemical agents.

* 1.2 Classification. The footwear cover shall be one type and shall be available in two sizes, small and large (see 6.2). The covers are the same for right and left feet.

Size small	-	Intended to be worn over sizes 3 through 7-1/2 all widths, combat boots
Size large	-	Intended to be worn over sizes 8 through 14 all widths, combat boots.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-L-61	-	Laces, Nylon
ZZ-G-381	-	Gloves, Rubber, Industrial
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8430

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MILITARY

MIL-D-12468 - Decontaminating Agents, STB

STANDARDS

FEDERAL

FED-STD-311 - Leather, Methods of Sampling and Testing
FED-STD-601 - Rubber: Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance Test Methods
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT LABORATORIES

8-1-365 - Footwear Cover, Chemical Protective (Overboots) Size Large
8-1-441 - Footwear Cover, Chemical Protective (Overboots) Size Small

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 395 - Rubber Property-Compression Set
D 412 - Rubber Properties in Tension
D 573 - Rubber Deterioration in an Air Oven
D 624 - Rubber Property-Tear
D 1053 - Measuring Rubber Property-Stiffening at Low Temperature Using a Torsional Wire Apparatus
D 2240 - Rubber Property-Durometer Hardness

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Samples.

3.1.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

3.1.2 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3 and 6.4).

3.2 Material (see 6.7).

* 3.2.1 Rubber compound. The upper portion of the cover shall be made of at least two plies of butyl rubber (see 4.4.1.1), pigmented black, suitably compounded and vulcanized together to meet the requirements of table I when tested as specified in 4.4.3.2. The soles and various rubber components of the cover shall be made of the same butyl compound and shall conform to the requirements of table I when tested as specified in 4.4.3.2.

TABLE I. Physical requirements of footwear cover

	Requirement	
	Soles	Uppers
Tensile strength, psi, minimum:		
As received	1400	1000
After aging	1200	900
After decontamination solution immersion	1000	700
Ultimate elongation, percent, minimum:		
As received	600	400
After aging	500	350
After decontamination solution immersion	450	300
Tear strength, pounds per inch, minimum	250	150
Hardness (Shore A Durometer)	60 ± 5	-
Compression set at 70°C for 22 hours, percent, maximum	50.0	-
Mustard resistance (break time in minutes), minimum	-	360
GB resistance (break time in minutes), minimum	-	450
Cold resistance	Pass 1/	-
Deformation due to decontamination solution, maximum:		
After immersion, wet, percent increase:		
- in thickness	50.0	50.0
- in length	15.0	15.0

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TABLE I. Physical requirements of footwear cover (cont'd)

	Requirement	
	Soles	Uppers
After immersion and aeration, percent increase:		
- in thickness	6.0	6.0
- in length	4.0	4.0
Porosity	-	Pass 2/
Thickness, inch, minimum	3/	0.025
Bond strength (pounds) minimum	18.0 4/	18.0

1/ Requirement value detailed in ASTM D 1053.

2/ There shall be no sign of porosity.

3/ Thickness to be as specified in Drawings 8-1-365 and 8-1-441.

4/ At no time during test separation shall the bond strength fall below 13.0 pounds.

* 3.2.2 Lace. The laces shall be 7/32 inch width, flat, spun nylon laces conforming to type I, class 2 of V-L-61. The length of each lace shall be 7 feet ± 2 inches. The color of the lace shall be black except the lace tips may be clear.

* 3.2.3 Instruction sheet. A paper instruction sheet, to include size marking, shall be placed inside each unit package of footwear covers so that the nomenclature marking and the size marking are readily visible. The size of the characters shall be 1/4-inch (18 point) for headings and size markings, and 1/8-inch (10 point) for wearing and donning instructions. The instruction sheet shall contain the information shown in figure 1 and the appropriate size marking as described in 3.5.3.2. The size marking shall be stamped or printed in black, after the nomenclature.

* 3.3 Design. The footwear cover available in two sizes shall be a loose fitting butyl rubber upper of at least two plies united together to form one ply, vulcanized to a non-slip molded butyl rubber sole with five holes to allow lacing around the foot (see figure 1). The compression molded sole shall conform to the requirements of Drawings 8-1-365 and 8-1-441. The calendered sole shall conform to Drawings 8-1-365 and 8-1-441, except that the cleats may be either round or oval. If oval, the minimum dimension of the width shall be 5/16 inch and the maximum dimension of the length shall be 7/16 inch. The calendered sole, except for the cleats, may be textured or pebbled. The reinforcing rings around the holes shall be molded in place on the compression molded sole. Over the calendered soles the rings shall be cut from an uncured sole compound and vulcanized to the sole so that the rubber becomes an integral part of the sole that cannot be forcibly removed without tearing the rubber. The cover is designed to be worn on either the right or left foot.

3.4 Color and finish. The footwear cover shall be black and shall have a smooth finish. A shallow cloth marking on the upper is permitted.

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hose of 3.3.5 to meet certain

6.3 Eyelets. Eyelet materials are restrictive to t

3.5 Construction.

* 3.5.1 Sole, upper, and weld strip. The soles, uppers, and weld strips shall be constructed in conformance with Drawings 8-1-365 and 8-1-441. The upper shall be bonded to the sole using strips of partially cured butyl rubber as the welding material. The upper shall be formed by bonding the front and rear edges using strips of partially cured butyl rubber as the welding material. The entire surface areas within the minimum dimensions (2-1/4 inch by 9-1/4 inch) of the sole weld shall be adhered to the upper and the sole.

* 3.5.1.1 Alternative construction. As an alternative, the construction of the footwear covers may be made using compression molded butyl uppers and seams. The butyl rubber upper covers will be made from uncured butyl by compression molding the seams at the same time and under the same conditions as the upper material is vulcanized. Welding strips will not be required if the upper and mating surfaces at the seams are cured at the same time under identical conditions. If required by the molding process, a 1/4-inch folded seam along the center line of symmetry on the upper illustrated on sheet 3 of Drawings 8-1-365 and 8-1-441 is acceptable. Welding strips will be required to bond the upper to the sole.

* 3.5.2 Vulcanization. The footwear covers shall be properly vulcanized. The seams shall be bonded so as to provide a waterproof barrier which does not separate.

* 3.5.3 Marking.

* 3.5.3.1 Identification marking. Each footwear cover shall be permanently and legibly marked in bold-faced Gothic capital letters not less than 1/4 inch in height with either yellow or white indelible marking ink (see 6.5). The marking shall remain clear and legible when tested as specified in 4.4.3.2. The inside of each cover shall be marked with the following legend (marking shall be centered on the side of the cover approximately 1 inch from the top of the cover):

NOMENCLATURE
SIZE
SPECIFICATION NO.
CONTRACT NO.
NATIONAL STOCK NO.
MANUFACTURER
DATE MANUFACTURED

* 3.5.3.2 Size marking. The outside of each footwear cover shall be marked with the appropriate size "SMALL (3 through 7-1/2)" or "LARGE (8 through 14)", conforming to the requirements of 3.5.3.1. The marking shall be centered on the outside of the side of the cover approximately 1 inch from the top of the cover.

* 3.5.4 Dusting. The inside and outside of each footwear cover shall be lightly dusted so as to produce a uniform dull appearance with whiting, talc, or other finely divided mineral matter which does not support mildew growth. After the covers have been tested for porosity, they shall be thoroughly dried and then dusted.

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* 3.5.5 Replacement of defective components. During the cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3.1 shall be removed from production and replaced with nondefective and properly matched components.

3.6 Workmanship. The finished footwear cover shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3.1 and tested for the characteristics specified in table II. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.4.1.1 Certification. The contractor shall furnish a certificate of compliance stating that the upper portion of the cover is made of at least two plies of butyl rubber as specified in 3.2.1.

* 4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are accomplished as specified and that component and assembly dimensions are as specified on the drawing. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

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4.4.3 End item inspection.

* 4.4.3.1 End item examination. The end item shall be examined for the defects listed below. The lot shall be expressed in units of one footwear cover. The sample unit shall be one footwear cover and selection shall be in pairs. The inspection level shall be level II. The acceptable quality levels (AQLs) expressed in defects per hundred units shall be as follows:

Critical defects	*
Major defects	1.0
Total defects (major and minor combined)	4.0

*NOTE Any critical defect found in sampling inspection for major and total defects shall be cause for rejection of the lot. One hundred percent inspection shall be performed for critical defects on each lot found acceptable under sampling inspection, and any footwear cover found to contain a critical defect shall be rejected.

Examine	Defect	Classification Critical	Major	Minor
Material	Any cut, tear, hole, or rupture in upper material		X	
	Any closed blister in upper material		X	
	Any burned spot which cracks on flexing, bending or stretching (by hand)		X	
	Any weak area such as: Pinch, pit, void, thin spot, abraided area, crease, or similar defective condition which results in a thickness at the defect less than the minimum required thickness	X		
	Any pit, void, or similar defective condition caused by the removal of readily removable foreign matter (see note), which results in a thickness at the defect less than the minimum required thickness		X	

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Examine	Defect	Classification Critical	Major	Minor
Material (cont'd)	More than three of the following defects of like description or more than three of the following defects in any combination, located within any one square inch area, but not resulting in a thickness at the defect less than the minimum required thickness:			
	Pinch, pit, void, abraided area, crease, pit or void caused by removal of readily removable foreign matter, (see note) or similar defective condition			X
	Any foreign matter not readily removable: resulting in lump exceeding 1/16 inch in length			X
	Any cut, tear or hole through soling material	X		
	NOTE: Readily removable foreign matter is defined as material which can be removed by flexing or stretching the cover by hand or by rubbing the defective area with the ball of the thumb or fingers. Cutting, scratching or otherwise puncturing the surface is not permitted.			
Design	Any characteristic not in accordance with specified requirements (i.e., any one of the five holes missing or improperly formed on rubber sole)		X	
Dusting	Missing or imcomplete (i.e., appearance not uniformly dull)			X
	Excessive (i.e., significant free dusting powder in polyethylene bag)			X
Color	Not black			X
Finish	Not smooth			X

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Examine	Defect	Classification Critical	Major	Minor
Laces	Missing			X
	Not tipped			X
Construction and workmanship	Component or assembly malformed to such a degree that:			
	-Serviceability is seriously affected		X	
	-Serviceability is affected but not seriously			X
	Tackiness (after dusting, see 3.5.4)	X		
	Opening of seam at top:			
	- Up to 1/4 inch			X
	- Over 1/4 inch		X	
	- Over 2 inches	X		
	Any opening on other parts of seam	X		
	Any repair or patch	X		
	Any area of sole weld strip within the minimum dimensions (2-1/4 by 9-1/4 inches) not adhered to upper sole		X	
Marking	Missing, incomplete, incorrect, not legible, not in proper location, or not accomplished in the specified manner			X

* 4.4.3.2 Testing of the end item. Each lot shall be tested for the characteristics shown in table II. Lot sizes shall be expressed in units of one footwear cover. All requirements shall be applicable to the sample unit. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in table II, they shall be as required by the referenced test method. All test reports shall contain the individual values used in expressing the final result.

a. Government acceptance tests (see 6.6). The Government shall perform acceptance tests for mustard and GB resistance. The sample unit shall be two footwear covers, and the inspection level shall be S-2. These tests are considered critical and any failure shall be cause for rejection of the lot represented by the sample.

b. Porosity test. The maximum lot size for the porosity test shall be 20,000 pair of footwear covers. The sample unit shall be one footwear cover, and the sample size shall be 800 individual footwear covers. Any footwear cover found to leak will be cause for rejection of the lot represented by the sample.

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c. All tests other than Government acceptance and porosity tests. For all remaining tests, the sample unit shall be twelve footwear covers. The sample size shall be as specified below. The lot shall be unacceptable if one or more sample units fail to meet any of the requirements specified.

Lot size	Sample size
800 or less	2
801 to 22,000 inclusive	3
22,001 and over	5

d. Government verification tests. The Government reserves the right to draw test samples from each lot to verify the contractor's test results, and to defer acceptance pending completion of Government verification testing.

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TABLE II. End item tests

Characteristic	Requirement paragraph	Test method	No. of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Material identification (rubber)	3.2.1	1/	1	X	
Tensile strength: 6/ 7/ 8/					
As received	3.2.1	D 412 2/ (method A)	5		10 psi
After aging	3.2.1	D 412 2/ 3/ (method A)	5		10 psi
After decontamination solution immersion	3.2.1	4.5.1	-		10 psi
Ultimate elongation: 6/					
As received	3.2.1	D 412 2/ (method A)	5		10 percent
After aging	3.2.1	D 412 2/ 3/ (method A)	5		10 percent
After decontamination solution immersion	3.2.1	4.5.1	-		10 percent
Tear strength 7/ 8/	3.2.1	D 624 2/ 4/	5		nearest pound per inch
Hardness	3.2.1	D 2240 2/	2		nearest whole number
Compression set	3.2.1	D 395 2/ (method B)	5		nearest percent
Mustard resistance	3.2.1	T209 or 204 5/	2		1 minute
GB resistance	3.2.1	T208 or 206 5/	2		1 minute
Cold resistance	3.2.1	4.5.2	3	X	

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TABLE II. End item tests (cont'd)

Characteristic	Requirement paragraph	Test method	No. of determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Deformation due to decontamination solution: 7/					
After immersion, wet:					
Thickness	3.2.1	4.5.3	3		0.1 percent
Length	3.2.1	4.5.3	3		0.1 percent
After immersion and aeration: 7/					
Thickness	3.2.1	4.5.3	3		0.1 percent
Length	3.2.1	4.5.3	3		0.1 percent
Porosity	3.2.1	4.5.4	1	X	
Thickness 7/	3.2.1	4.5.5	-	-	-
Bond strength	3.2.1	4.5.6	-	-	0.1 pounds
Reinforcing ring adhesion (for calendered sole)	3.3	4.5.7	1	X	
Permanence of marking	3.5.3.1 and 3.5.3.2	4.5.8	1	X	

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Refers to ASTM Standard.

3/ The specimen shall be aged at 212°F for 48 hours in accordance with ASTM D 573 before being tested for tensile strength and elongation.

4/ Die C shall be used.

5/ Refers to MIL-STD-282.

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6/ When testing the soles, the specimen shall be taken in the smooth area of the sole that is free of the tread design. If necessary the tread protrusion shall be buffed off the sole in order to provide an area of uniform thickness for obtaining the specimen.

7/ All thickness measurements for testing and calculation purposes shall be determined on the test specimen without buffing or abrading the surface fabric impression.

8/ Specimens shall be taken so that the lengthwise dimension is in the vertical (leg) direction of the footwear cover.

4.4.4 Packaging inspection. An examination shall be made to determine that the preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination for the closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, loose strapping, inadequate stapling, or improper taping. Open and non-continuous heat sealed seams and closure of polyethylene bag. Incorrectly fabricated bag. Bulged or distorted container.
Content	Number of footwear covers per container is more or less than required.

* 4.4.5 Examination for palletization. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

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Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Method of inspection.

4.5.1 Tensile strength and ultimate elongation after decontamination. The tensile strength and ultimate elongation shall be determined in accordance with ASTM D 412 (method A). The specimens shall be obtained as indicated in 4.5.1.1 and 4.5.1.2. The cross sectional area of the specimen used for tensile strength shall be calculated using the thickness determined in 4.5.3.a and 4.5.3.b.

4.5.1.1 Uppers. Three specimens shall be cut from each upper that was tested for decontamination. The specimens shall be cut from the 3 by 6 inch area marked on the uppers. The long dimension of the specimens shall be parallel to the long dimension of the 3 by 6 inch area. The specimens shall be cut in such a manner that the ends of the specimens are the same distance from the edges of the marked area.

4.5.1.2 Soles. One specimen shall be cut from the sole of each footwear cover that was tested for decontamination. The specimen shall be taken from the 1 by 4-1/2 inch area marked on the sole. The specimen shall be cut with its long dimension parallel to the long dimension of the marked area.

4.5.2 Cold resistance. The soles of three footwear covers from each sample unit shall be tested as minus 20° ± 2°F in accordance with the Routine Inspection and Acceptance paragraphs of ASTM D 1053. The coolant shall be dry ice cooled methenol and exposure time shall be 10 minutes.

4.5.3 Resistance to decontamination solution. Three footwear covers from each sample unit shall be tested. The resistance to decontaimination solution shall be determined as follows:

a. Accurately mark a 3 by 6 inch area on the upper of each cover. Draw three straight lines along the entire length of the area as follows: The first line shall be 1/2 inch in from one edge of the marked area, the second line 1/2 inch in from the opposite edge. The third line shall be 1-1/2 inches from both edges.

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Then draw three straight lines across the entire width of the area as follows: The first line shall be 2 inches in from one edge of the marked area, the second line 2 inches in from the opposite side. The third line shall be 3 inches from both edges. Using a micrometer as described in Method 2011 of FED-STD-601, determine the thickness of the area at the nine points where the three length lines cross the three width lines. The thickness of the specimen shall be the average of the nine values.

b. On a smooth surface of each sole, mark an area 1 inch in width by 4-1/2 inches in length. Using the micrometer described in Method 2011 of FED-STD-601, determine the thickness of the sole in three places spaced equidistant along the longitudinal center line of the marked area. The thickness shall be recorded as the average of the three values. If necessary, the tread protrusion shall be buffed off the sole in order to provide an area of uniform thickness.

c. Mount the footwear cover on a suitable form and immerse in test solution to within 1 inch of the top edge, taking care not to wet the interior. The test solution shall be Super-tropical bleach slurry. The slurry shall be made up (by weight) of 1 part Super-tropical bleach conforming to MIL-D-12468 and 2 parts water. (Note: Super-tropical bleach is corrosive to most metals and is injurious to most fabrics. A protective mask and gloves should be worn when handling this material.) The temperature of the slurry during the test shall be 80°F to 90°F. The time of immersion shall be 5 minutes.

d. After the 5 minute immersion in STB solution, withdraw the footwear cover, wash off the slurry with water, and rinse in 95 percent ethyl alcohol. While the footwear cover is still wet, measure the marked areas immediately for thickness and length. Thickness shall be determined as described in "a" and "b" above. The length shall be measured along the three length lines of the upper and recorded as the average of the three measurements. The length for the sole shall be measured along the longitudinal center line of the marked area.

e. Place the wet footwear covers on heavily talced paper towels and allow to stand at room temperature for 18 hours.

f. After the 18 hours air drying, remeasure the marked areas for thickness and length. Determine thickness in accordance with "a" and "b" above. Determine the length in accordance with "d" above.

g. The percent increase in thickness and percent increase in length shall be calculated as follows: 1/

$$\text{Percent increase in thickness after immersion (wet)} = \frac{(T_1 - T)\ 100}{T}$$

$$\text{Percent increase in thickness after immersion and aeration} = \frac{(T_2 - T)\ 100}{T}$$

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$$\text{Percent increase in length after immersion (wet)} = \frac{(L_1 - L)\ 100}{L}$$

$$\text{Percent increase in length after immersion and aeration} = \frac{(L_2 - L)\ 100}{L}$$

Where:

T = Original thickness as determined in "a" and "b" above as applicable.
T_1 = Thickness after immersion as determined in "d" above.
T_2 = Thickness after immersion and aeration as determined in "f" above.
L = Original length (6 inches or 4-1/2 inches as applicable).
L_1 = Length after immersion as determined in "d" above.
L_2 = Length after immersion and aeration as determined in "f" above.

1/ Separate determinations shall be made for the upper and the sole.

4.5.4 Porosity. The porosity test shall be performed as specified in ZZ-G-381 except the test apparatus shall be modified to accommodate the open end of the footwear cover and the air pressure shall be 0.5 pound per square inch.

4.5.5 Thickness. The thickness of the footwear cover shall be determined in accordance with Method 2011 of FED-STD-601. Three footwear covers from each sample unit shall be tested. Four measurements shall be made on the upper and four measurements on the sole. The sample unit shall fail if any single measurement is less than the minimum specified.

4.5.6 Bond strength. The apparatus described in Method 2031 of FED-STD-311 shall be used for testing the bond strength of the seams of the uppers and the bond strength between the sole and the upper. Five footwear covers from each sample unit shall be tested as indicated in 4.5.6.1 and 4.5.6.2.

4.5.6.1 Bond strength of upper seams. Two 1 by 4 inch specimens from each cover shall be tested. One specimen shall be cut from the front seam and one specimen shall be cut from the back seam. The specimens shall be cut with their long dimension perpendicular to the length of the seam. Each end of the specimen shall be fastened in the grips of the test machine and the machine operated until the seam is completely separated. The maximum value obtained shall be recorded as the bond strength.

* 4.5.6.2 Bond strength of sole to upper. A 1 by 5 inch specimen shall be cut from each footwear cover with the long dimension of the specimen parallel to the long dimension of the sole. Not less than a 4 inch portion of the specimen shall consist of the adhered upper and sole. The sole portion of the specimen shall be fastened in one grip of the machine and the upper portion fastened in the other grip. The machine shall be operated until 4 inches of the adhered area of the specimen has been separated. The maximum and minimum bond strengths obtained during separation shall be recorded.

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* 4.5.7 Reinforcing ring adhesion test (calendered sole). The rings shall be forcibly removed from the soles by use of pliers or other suitable hand tools. Any ring which is removed without tearing rubber from the ring or from the sole shall be considered a test failure.

4.5.8 Permanence of marking. The footwear cover shall be tested to determine the durability of the marking as follows:

a. Boil the footwear cover for 2-1/2 to 3 hours in a solution containing 1.25 ounce/gallon of standard calcium hypochlorite bleach powder (70 percent available chlorine).

b. Remove the footwear cover from the bleach solution and soak in fresh water at 70°F for 15 minutes. Change water and repeat this step two additional times.

c. Hang the footwear cover and allow to dry at about 110°F. When the footwear cover is thoroughly dry, examine for legibility of the marking.

* 5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

* 5.1.1 Level A. Each pair of footwear covers, with two pairs of laces, shall be banded together with the sole bottoms facing out. Excess material and laces shall be tucked inside the folds. Three rubber bands shall be used to secure the paired covers, with one band placed around the pair approximately 3 inches in from each end and the third placed approximately around the center. The banded pair of covers shall measure 19 inches in length and 4-3/4 inches at its widest point. Each pair of covers, with instruction sheet (see 3.2.3), shall be unit packed in a snug-fitting, flat style, clear polyethylene film bag of .0015 inch thickness (+ 25% tolerance). The bag shall be formed with heat sealed seams that are straight, continuous, and parallel to each other. The bag closure shall be heat sealed with the seal made as close as possible to the open end. The bag may be fabricated from polyethylene film tubing or sheeting. Prior to or during the closing operation, excess air within the bag shall be expelled.

* 5.1.2 Commercial. Footwear covers shall be preserved in accordance with MIL-STD-1188.

* 5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

*5.2.1 Level A packing. Twenty pairs of footwear covers of one size only, unit packed as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed

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flat, five in length, one in width, and four in depth within the shipping container. Inside dimensions of each shipping container shall be 23-1/4 inches in length, 18-3/4 inches in width, and 9-1/2 inches in depth. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Toward the end of the contract or when there are less than the required number of footwear covers per container of the same size, mixed sizes may be packed within the same container.

* 5.2.2 Level B packing. Twenty pair of footwear covers of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, five in length, one in width, and four in depth within the shipping container. Inside dimensions of each shipping container shall be 23-1/4 inches in length, 18-3/4 inches in width, and 9-1/2 inches in depth. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4. Toward the end of the contract or when there are less than the required number of footwear covers per container of the same size, mixed sizes may be packed within the same container.

* 5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

* 5.2.3 Commercial. Footwear covers, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

* 5.3 Palletization. When specified (see 6.2), footwear covers packed as specified in 5.2 shall be palletized in accordance with lead type Ia of MIL-STD-147. Each prepared load shall be bonded with primary anad secondary straps in accordance with bonding means K or L. Pallet pattern shall be in accordance with load pattern number 90 of the appendix of MIL-STD-147. Interlocking of loads shall be affected by reversing the pattern of each course.

* 5.4 Marking. In addition to any special marking required by the contract, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

* 5.4.1 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches in size, with the words "MIXED SIZES" plainly printed or stamped on it and under these words shall be legibly printed or stamped the correct quantity of pairs and sizes within the container.

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6. NOTES

6.1 Intended use. The footwear covers are intended for use over conventional footwear in conjunction with the Suit, Chemical Protective and Clothing Outfit, Chemical Protective.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) When a first article is required (see 3.1.2, 4.3, and 6.4).
(d) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(f) When palletization is required (see 5.3).

6.3 Sample. For access to samples, address the procuring activity issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The first article should consist of two footwear covers. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.5 Marking ink. Markem JXN-7410-H-White or Yellow, made by Markem Machine Company, 40 Putnam Street, Keene, NH 03431, has been found to be suitable when dried under infrared light for 5 minutes then air drying for 24 hours.

6.6 Government acceptance tests. Samples for Government acceptance tests (see 4.4.3.2a) should be forwarded to Commander, US Army Armament Research and Development Command, ATTN: Test Br/Bldg E5100, Aberdeen Proving Ground, MD 21010.

6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

6.8 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a covenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - EA
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DLA - CT

User activities:

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Preparing activity:

Army - GL

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FOOTWEAR COVER, CHEMICAL PROTECTIVE, (OVERBOOTS)

WEARING INSTRUCTIONS

1. THESE FOOTWEAR COVERS ARE TO BE WORN OVER CONVENTIONAL FOOTWEAR IN THE EVENT CHEMICAL PROTECTION (CB PROTECTION) IS REQUIRED. FOOTWEAR COVERS SIZE SMALL SHALL BE WORN OVER SIZES 3 THRU 7½ COMBAT BOOTS, ALL WIDTHS. FOOTWEAR COVERS SIZE LARGE SHALL BE WORN OVER SIZE 8 THRU 14 COMBAT BOOTS ALL WIDTHS.

2. ALWAYS WEAR THE FOOTWEAR COVERS UNDER YOUR OUTER TROUSERS.

3. PERIODICALLY, INSPECT YOUR CHEMICAL PROTECTIVE FOOTWEAR COVERS FOR HOLES OR PUNCTURES. IN THE EVENT THEY DO DEVELOP HOLES, ARE PUNCTURED OR BECOME TORN, DISCARD IMMEDIATELY AND USE A SERVICEABLE PAIR.

4. FOOTWEAR COVERS EXPOSED TO TOXICOLOGICAL AGENTS MUST BE DECONTAMINATED BEFORE RE-USE. THE FOOTWEAR COVERS CAN BE DECONTAMINATED BY USING THE PROCEDURES SPECIFIED IN TM 3-220; CHEMICAL, BIOLOGICAL, AND RADIOLOGICAL (CBR) DECONTAMINATION.

5. SHOULD YOUR FOOTWEAR COVERS BECOME CONTAMINATED WITH GASOLINE, OIL, GREASE OR CLEANING FLUIDS, WIPE OFF AND AIR DRY WITHIN TWO MINUTES.

DONNING INSTRUCTIONS

A. PUT THE OVERBOOT ON THE FOOT AND POSITION IT SO THAT THE FOOT IS CENTRAL ON THE OVERBOOT SOLE, BOTH FROM SIDE TO SIDE AND FRONT TO BACK. TUCK FRONT EXCESS OF UPPER UNDER FRONT OF FOOT.

B. THREAD THE LACES THROUGH THE FRONT EYELET SO THAT BOTH ENDS ARE OF THE SAME LENGTH. PASS THE LACE ENDS THROUGH THE SIDE EYELET FROM INSIDE TO OUTSIDE AND DRAW UP TIGHTLY, FASTENING THE LACE WITH A SINGLE KNOT (DIAGRAM 1).

C. PASS THE ENDS THROUGH THE REAR EYELETS FROM INSIDE TO OUTSIDE AND PULL UP FIRMLY (DIAGRAM 2). POSITION FISH-TAIL NEATLY AT SIDES OF ANKLE AND TIE FIRMLY WITH A BOW (DIAGRAM 3), IN FRONT.

D. TUCK IN THE LOOSE ENDS TO NEATEN THE LACES. THE SPARE MATERIAL OF THE UPPERS SHOULD NOW BE PULLED AWAY FROM THE EXTREMITIES TO TIDY THE TIED OVERBOOTS.

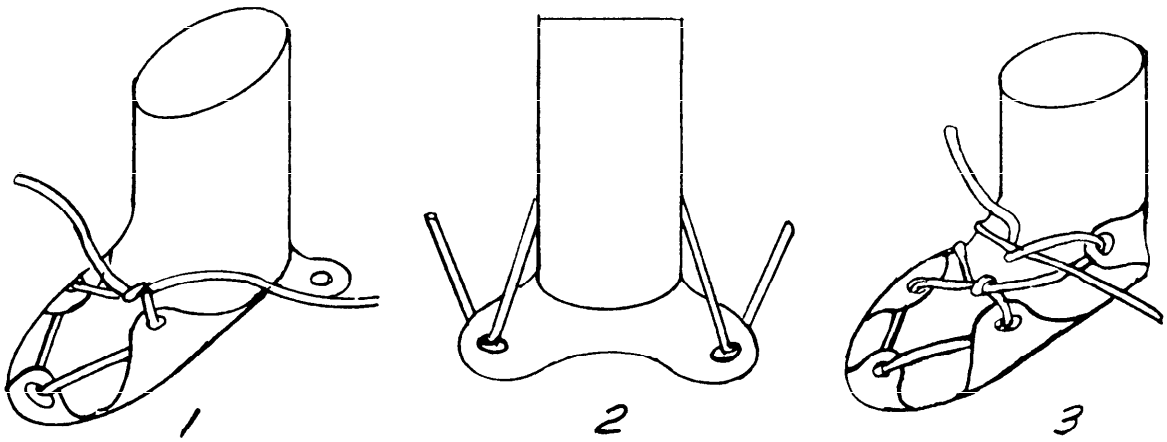


FIGURE 1

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-43987A Footwear Cover, Chemical Protective (Overboots)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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